Experiment 11-Creating a Pandas DataFrame with Random Values and Highlighting NaN (Missing) Values

Aim

To create a DataFrame with random values, replace selected values with NaN, and highlight the missing values using Pandas Styler.

Procedure

1. Import the Pandas and NumPy libraries.
2. Create a DataFrame with random values.
3. Replace selected values with NaN.
4. Define a function to identify NaN values.
5. Use `style.map()` to highlight the NaN values in red.
6. Display the styled DataFrame.

PROGRAM

```
# Import libraries
import pandas as pd
import numpy as np

# Create DataFrame with random values
np.random.seed(10)    # For same output every time
df = pd.DataFrame(np.random.randn(10, 4),
                  columns=['A', 'B', 'C', 'D'])

# Convert some values to NaN
df.iloc[0, 2] = np.nan
df.iloc[3, 3] = np.nan
df.iloc[4, 0] = np.nan
df.iloc[9, 3] = np.nan

# Function to highlight NaN values
def highlight_nan(val):
    if pd.isna(val):
        return 'background-color: red'
    return ''

# Display highlighted DataFrame
styled_df = df.style.map(highlight_nan)

print(df)
styled_df = df.style.map(highlight_nan)

styled_df.to_html("output.html")

print("Output saved as output.html")
```

OUTPUT

	A	B	C	D
0	1.331587	0.715279	nan	-0.008384
1	0.621336	-0.720086	0.265512	0.108549
2	0.004291	-0.174600	0.433026	1.203037
3	-0.965066	1.028274	0.228630	nan
4	nan	0.135137	1.484537	-1.079805
5	-1.977728	-1.743372	0.266070	2.384967
6	1.123691	1.672622	0.099149	1.397996
7	-0.271248	0.613204	-0.267317	-0.549309
8	0.132708	-0.476142	1.308473	0.195013
9	0.400210	-0.337632	1.256472	nan

Experiment 12-Creating a Pandas DataFrame with Random Values and Setting Background Color to Black and Font Color to Yellow

Aim

To create a DataFrame with 10 rows and 4 columns containing random values and apply black background with yellow font using Pandas Styler.

Procedure

1. Import the Pandas and NumPy libraries.
2. Create a DataFrame with random values.
3. Use `set_properties()` to set the background color to black.
4. Set the font color to yellow.
5. Display the styled DataFrame or save it as an HTML file.

PROGRAM

```
import pandas as pd
import numpy as np

# Create DataFrame
np.random.seed(10)
df = pd.DataFrame(np.random.randn(10,4),
                  columns=['A','B','C','D'])

# Apply black background and yellow text
styled_df = df.style.set_properties(
    **{
        'background-color': 'black',
        'color': 'yellow'
    }
)

# Display
styled_df.to_html("output.html")
print("Output saved as output.html")
```

OUTPUT

	A	B	C	D
0	1.331587	0.715279	-1.545400	-0.008384
1	0.621336	-0.720086	0.265512	0.108549
2	0.004291	-0.174600	0.433026	1.203037
3	-0.965066	1.028274	0.228630	0.445138
4	-1.136602	0.135137	1.484537	-1.079805
5	-1.977728	-1.743372	0.266070	2.384967
6	1.123691	1.672622	0.099149	1.397996
7	-0.271248	0.613204	-0.267317	-0.549309
8	0.132708	-0.476142	1.308473	0.195013
9	0.400210	-0.337632	1.256472	-0.731970